COMPUTER NETWORKS

CSE 321

Unique Security Challenges in IoT

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Unique Security Challenges in IoT

- Resource Constraints – Low memory, Low energy
- Heterogeneity – Device, protocol and platform diversity
- Scalability Issues – Affects monitoring and security
- Physical Accessibility – Public spaces or remote locations

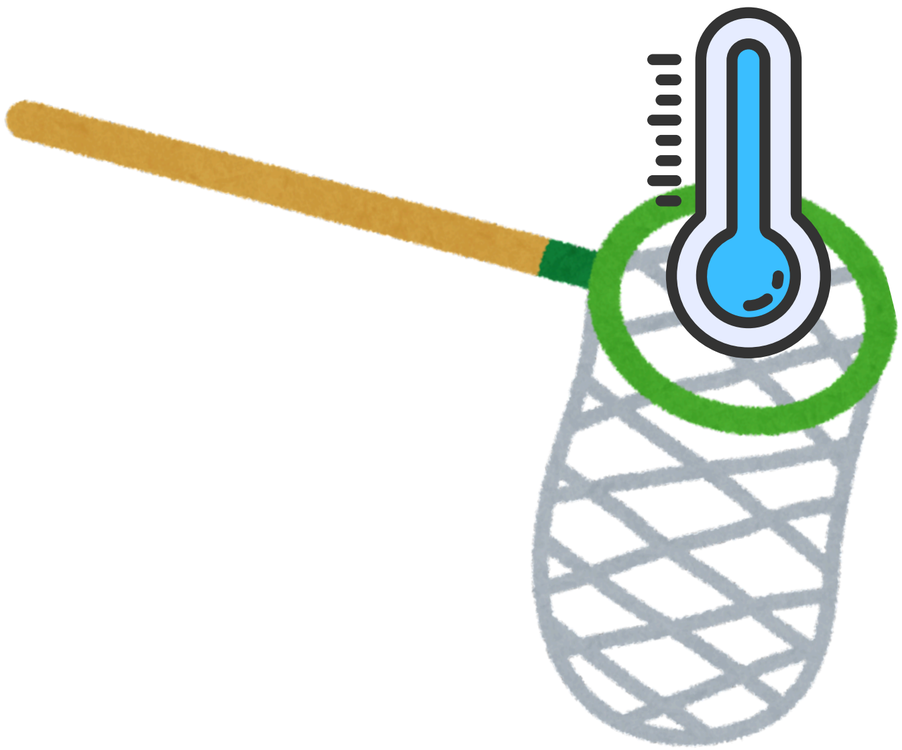
Some Common Type of Attacks

Physical Layer Attacks

Physical Layer Attacks

Node Tampering and Capturing

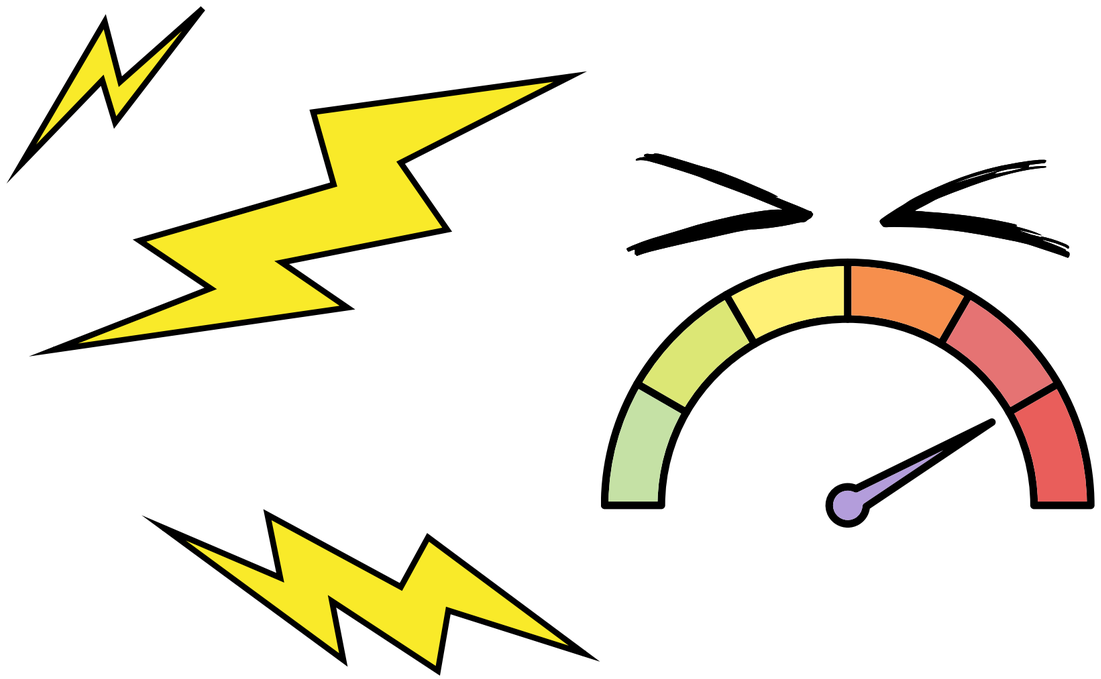
Physical manipulation to extract information or modify functionality



Physical Layer Attacks

Environmental Attacks

exploitation of physical deployment environments to disrupt functionality (radio frequency jamming, electromagnetic interfarence etc)



Network Layer Attacks

Network Layer Attacks

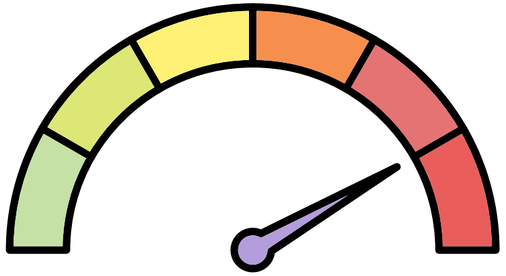
DoS Attacks

Denial of Service attacks prevent legitimate users from accessing services.

Network Layer Attacks

Eavesdropping

Eavesdropping involves passive interception to collect sensitive information



Case Study

Mirai Botnet

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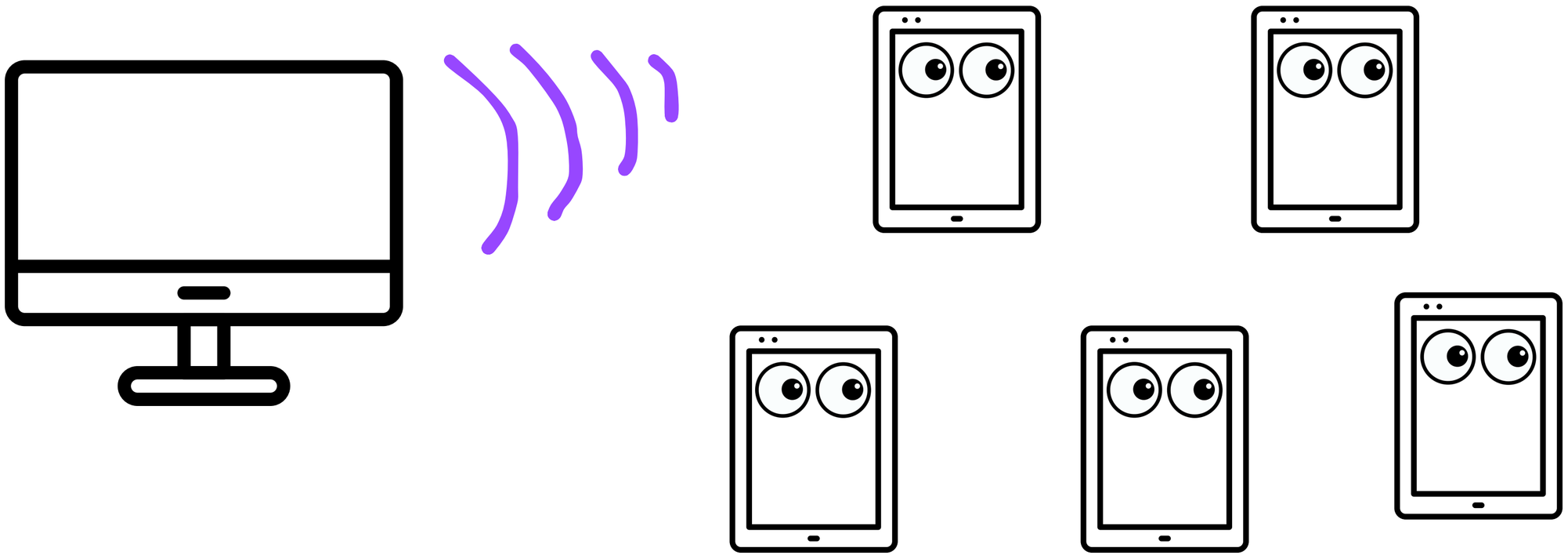
Mirai Botnet

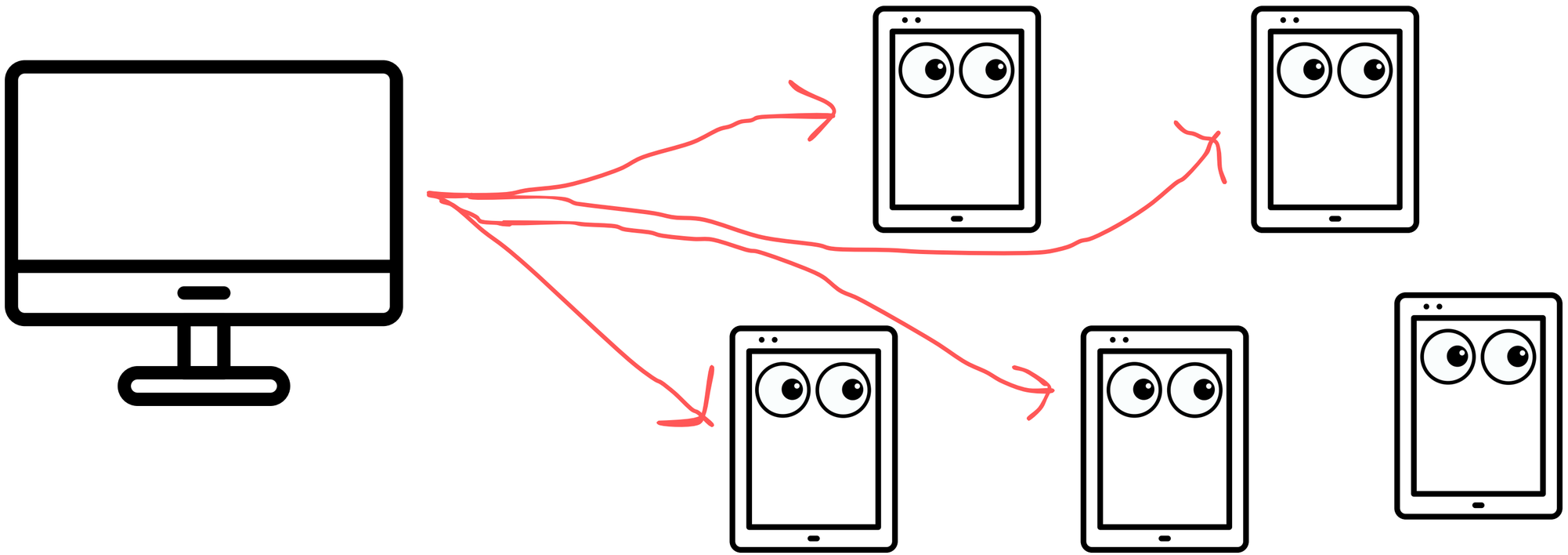
Mirai scanned for IoT devices with open Telnet ports, attempting authentication using default credentials



Case Study

Mirai Botnet

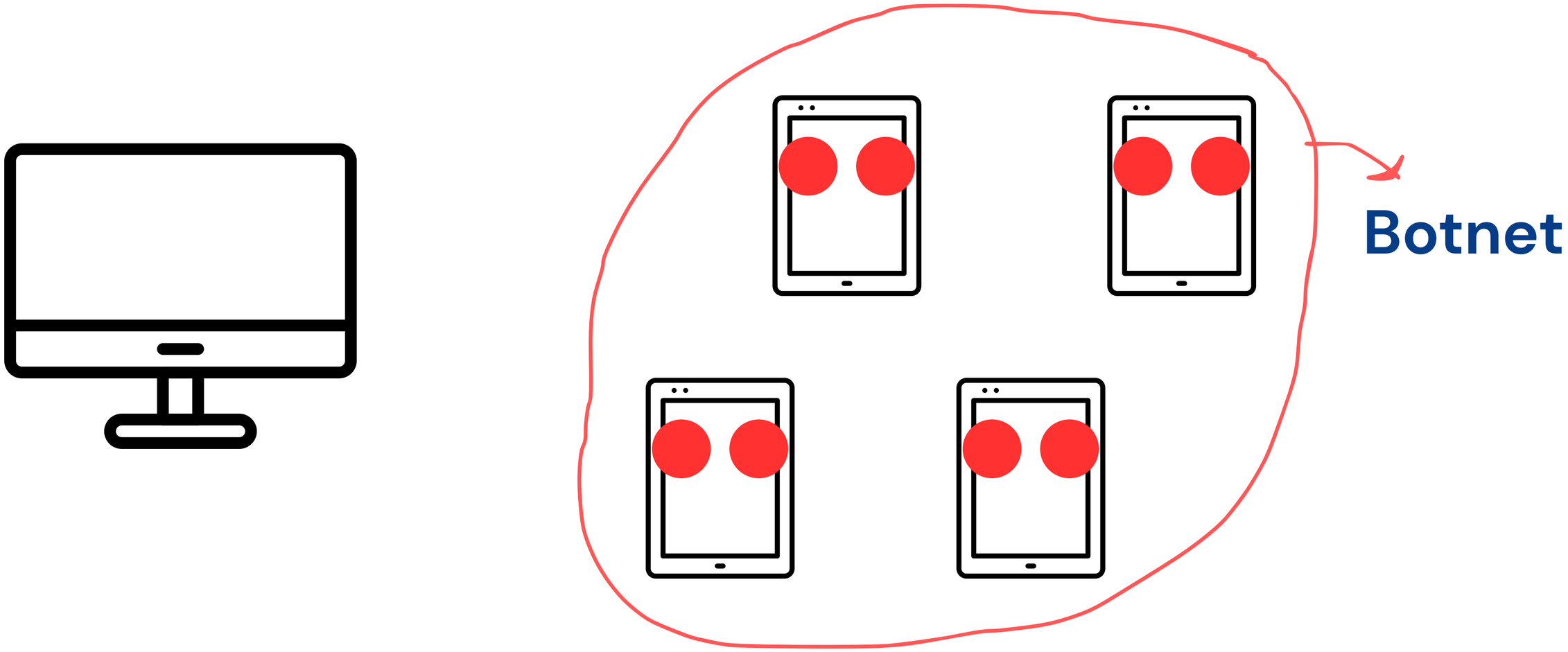
Compromised devices installed malware connecting to command-and-control infrastructure



Case Study

Mirai Botnet

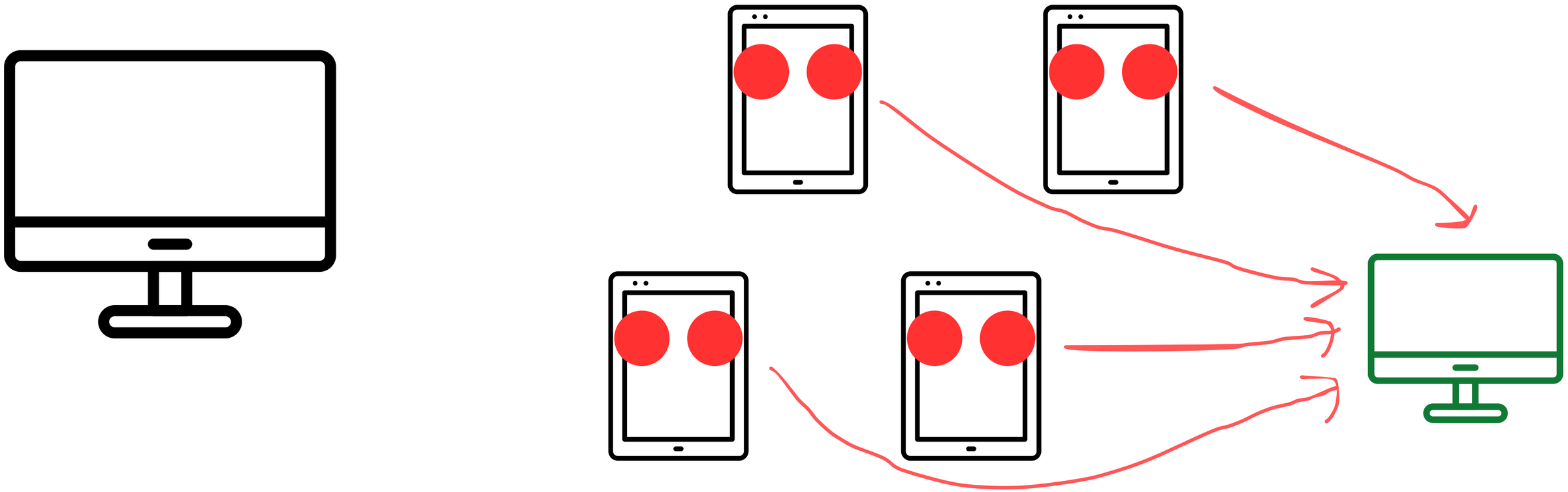
At peak, Mirai comprised hundreds of thousands of devices



Case Study

Mirai Botnet

Mirai launched attacks including massive DDoS against DNS provider Dyn, disrupting major Internet services and generating traffic exceeding 1 Terabit per second



Defense Mechanisms

- Cryptographic Solutions

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- Authentication and Access Control

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- Intrusion Detection and Prevention Systems
- Secure Communication Protocols

Defense Mechanisms

- Cryptographic Solutions
- Authentication and Access Control
- Intrusion Detection and Prevention Systems
- Secure Communication Protocols
- Physical Security

Resource

A Comprehensive Survey on Cybersecurity Threats and Defense Mechanisms in IoT Networks

OUR DEVICES ARE NOW 100% SECURE.
I TURNED THEM ALL OFF.
HOW DID YOU DO THAT?
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